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The Promise of Precision Nutrition for Modulation of the Gut Microbiota as a Novel Therapeutic Approach to Acute Graft-versus-host Disease

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Abstract. Acute graft-versus-host disease (aGVHD) is a severe side effect of allogeneic hematopoietic stem cell transplantation (aHSCT) that has complex phenotypes and often unpredictable outcomes. The current management is not always able to prevent aGVHD. A neglected actor in the management of aGVHD is the gut microbiota. Gut microbiota dysbiosis after aHSCT is caused by many factors and may contribute to the development of aGVHD. Diet and nutritional status modify the gut microbiota and a wide range of products are now available to manipulate the gut microbiota (pro-, pre-, and postbiotics). New investigations are testing the effect of probiotics and nutritional supplements in both animal models and human studies, with encouraging results. In this review, we summarize the most recent literature about the probiotics and nutritional factors able to modulate the gut microbiota and we discuss the future perspective in developing new integrative therapeutic approaches to reducing the risk of graft-versus-host disease in patients undergoing aHSCT.

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INTRODUCTION

Allogeneic hematopoietic stem cell transplantation (aHSCT) is applied as first-line therapy in a wide variety of severe immunodeficiencies and bone marrow failures or as definitive treatment in high-risk hematologic malignancies. Today, aHSCT is a standard clinical practice in hundreds of specialized clinical centers, serving as a potentially lifesaving treatment for tens of thousands of patients every year, throughout the world.

A major limitation of its application is the graft-versus-host disease (GVHD) complication, occurring in acute GVHD (aGVHD) involving gastrointestinal (GI) tract, liver, and skin in 35%–50% of cases and in chronic GVHD (cGVHD) in 30%–40% of cases.[1,2] Furthermore, the outcomes of aGVHD vary unpredictably between the mild and severe forms. Thus, at present, the clinical outcome for these patients is dismal, with long-term morbidity in 10%–50% of adult aGVHD cases and 50%–70% of pediatric cases.

For many decades now, aGVHD has been a focus of intense research designed to discover new biomarkers and therapeutic targets to improve clinical outcomes. Interestingly, one of the most recent important discoveries focused on 2 GI biomarkers: suppressor of tumorigenesis 2 (ST2) and regenerating islet-derived 3 alpha (REG3α). Both molecules, combined in an algorithm, predicted GVHD severity and treatment response, achieving therefore an exceptional performance.[3] The GI tract is indeed a key component in the biology of aGVHD, representing an interphase between the gut lumen microbiota composition[4] and epithelium-associated immune tissues. As such, the GI tract is the first mediator of inflammatory signals, typically in the case of chemo/radiotherapy damages after aHSCT conditioning. ST2 and Reg3α are both biomarkers of GI crypt damage, proving the key importance of the disruption of GI barriers, and implying the relevance of finding biological targets for drugs in this ecosystem.

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GI symptoms are common in aGVHD patients, causing great discomfort for the patients and impacting their nutritional status, with vomit, diarrhea, loss of appetite, and, consequently, loss of weight. It is described how a massive microbial shift occurs after the transplant due to conditioning treatments and slowly recovers to balance.[5,6] Most importantly, also patient nutritional status and dietary intake have an impact on the microbiota composition,[7] and they can be targeted to modulate and reestablish a healthy gut microbiota.

In this review, we focus on probiotics and nutritional supplements as a new therapeutic approach to restore the microbial flora and potentially prevent or revert aGVHD. Such an approach falls under the new concept of precision nutrition that aims to adopt comprehensive nutritional recommendations based on multiple individual factors, such as genetics, dietary habits, lifestyle, and metabolome,[8-10] all strongly linked to the gut microbiota.[11] Precision nutrition is a promising approach able to improve the patients' outcomes and quality of life, with minimal risks, minimal cost, and great potential benefit, for any patient undergoing aHSCT. Therefore, this review poses an innovative point of view about the role of nutrition supplements and probiotics in modulating the gut microbiota, a topic that is getting great interest from the experts working in the field of aGVHD.

NUTRITIONAL STATUS IN aGVHD

The nutritional status of aHSCT patients is a critical factor in clinical practice; however, the heterogeneity of the clinical symptoms of aGVHD and defining the optimal nutritional guidelines remain a challenge.

Both malnutrition and obesity are risk factors for morbidity and mortality in aHSCT patients. Malnutrition, in particular undernutrition, is defined as a negative nutritional imbalance due to inadequate intake of calories and nutrients. Malnutrition is frequently associated with chronic disorders and is diagnosed with the presence of at least 2 of the following indices: inadequate energy intake; weight loss, characterized mainly by a reduction in muscle and fat masses; fluid accumulation; and reduced grip strength.[12] The causes of malnutrition in aHSCT patients are multiple, before, during, and after the transplant. Frequently, patients suffer from nausea, mucositis, vomiting, and diarrhea caused by the conditioning regimen and symptoms related to infectious complications or supporting therapy.[13-15] Pretransplant underweight increases the risk of aGVHD, and it should require an accurate nutritional management of the patient.[16] After transplantation, it is recommended that patients undergo comprehensive nutritional screening to optimize the nutritional support and reduce the risk of malnutrition and GI symptoms.[12] The onset of GVHD after transplantation can exacerbate the GI symptoms, leading to malabsorption, weight loss, dehydration, and electrolyte loss.[17] With some exceptions, enteral nutrition (EN) is preferred to parenteral nutrition (PN) in the most recent practice with the aim of improving the mucosal repair, decreasing the incidence of hyperglycemia and infection and restoring the balance of the gut microbiota and it is recommended in the current guidelines.[18,19]

Obesity shows controversial effects on aHSCT outcomes. Obesity is a recognized risk factor in the most clinically relevant (HCT)-specific comorbidity index, commonly known as the "Sorror index."[20] A human study conducted on the role of the pretransplant BMI in aHSCT showed an increased risk of GVHD and higher no-relapse mortality in overweight and obese subjects.[15] Another similar study confirmed an increased no-relapse mortality in overweight and obesity, wherein the major cause of relapse was GVHD, a lower incidence relapse, and a higher level of the inflammatory biomarkers ST2 and tumor necrosis factor receptor 1.[21] Studies performed in animal models confirmed that diet-induced obesity increases the rates of aGVHD with around a 5-fold increase in mortality for a BMI >30 kg/m^2 before transplantation.[22] Interestingly, the same team discovered that aGVHD is driven more by an obesity phenotype than by a high-fat diet resulting in higher aGVHD incidence and higher mortality.[23] Opposite findings were shown by Voshtina et al that comparing elderly patients with BMI >30 kg/m^2 versus BMI <30 kg/m^2 did not find any significant difference in the incidence of aGVHD and cGVHD and a not-significative trend for a higher overall survival (OS) with BMI <30 kg/m^2.[24] Similarly, other studies confirmed no association of BMI with GVHD, infections and OS, but with shorter engraftment time[25] and higher OS.[26] A big study cohort from the Japanese marrow donor program showed a higher risk of aGVHD grade II-IV associated with pretransplant BMI >30 kg/m^2 and confirmed that obesity increases the risk of systemic infections.[27] A different effect of body weight has been shown between transplants from related donors (RDs) and unrelated donors (URDs) with increased mortality for underweight patients in RDs and for obese subjects in URD transplants, and higher incidence of aGVHD in obese RD-transplanted patients, even though the authors concluded that underweight poses higher mortality risk after transplant, whether obesity does not preclude safety and effectiveness of hematopoietic stem cell transplantation (HSCT).[28] The discordant results are probably due to different investigation protocols applied to different types of cohorts. Further investigations need to clarify the best approach to body weight management for HSCT patients.

Deficiencies in the serum level of minerals and vitamins are commonly caused by nutritional deficits. Recent evidence suggests the importance of an adequate intake of micronutrients to reduce GVHD comorbidities, such as muscle mass loss, ocular manifestations, and osteoporosis, with adequate intake of vitamins A and D, zinc, magnesium and potassium, omega-3 fatty acids, and probiotics found to be of particular importance.[29] Specific nutrients are described as immune system modulators if supplemented in the nutritional therapy after transplantation. These supplements include glutamine to improve enterocyte metabolism, eicosapentaenoic acid to reduce levels of the proinflammatory cytokines, interleukin 6 and leukotriene B4, N-acetylcysteine to protect hepatocytes and scavenge free radicals, and selenium for its antioxidant properties and to reduce levels of interleukins and tumor necrosis factor. The immune-nutritional activities of these supplements have been harnessed to reduce the risk of GVHD, but do not protect against infections.[30] In a study of its effects on human peripheral blood mononuclear cells, polyphenolic extract from extra virgin olive oil (PE-EVOO) showed immune-modulatory capacity and diminished the activation of T cells and their cytokines.[31] The same study confirmed the anti-inflammatory activity of PE in a murine model of bone marrow transplantation by demonstrating a reduction in proinflammatory cytokine levels at 5 d after transplantation, as well as a decrease in the occurrence of GVHD. Furthermore, these effects were accompanied by improved intestinal histopathology and the restoration of butyrate levels in mice treated with PE-EVOO.[31]

Overall, the current evidence indicates that the correct nutritional support is an important factor in reducing the occurrence of aGVHD, and most importantly, this approach can reduce gut dysbiosis, defined as the "disruption in the normal intestinal colonization,"[32] in patients after aHSCT.

THE HUMAN GUT MICROBIOTA AND aGVHD

In the last decade, the human gut microbiota have attracted great attention from researchers due to their potential regulatory roles in pathophysiology. Strong evidence suggests the involvement of the microbiome in a broad range of pathologies, including not only nutrition-related diseases such as obesity, diabetes, and malnutrition,[33] but also immunological diseases, such as inflammatory bowel disease, and aGVHD.[34-37] The definition of the role of each microbe as part of the human microbiota is a major challenge, since it contains approximately 10–100 trillion symbiotic microbes, including bacteria, fungi, parasites, and viruses. A healthy gut microbiota depends on the maintenance of the balance between "good" and "bad" bacteria, and anything that disrupts this balanced composition is defined as a "manipulator" of the gut microbiota. Healthy gut microbiota also depend on their level of diversity, with higher diversity associated with a healthier gut microbiota.[38-40] The main biological functions of the microbiota are metabolism, immune system regulation, and defense against exogenous pathogens[41-44] through a variety of mechanisms, such as fermentation of indigestible substrates and endogenous intestinal mucus, the production of short-chain fatty acids (SCFAs) and gases, energy extraction from the diet, fatty acid break-down and storage in the liver and adipose tissue, uptake and synthesis of vitamins and minerals, regulation of intestinal permeability and secretion of enteroendocrine hormones, bile acid metabolism, and regulation of metabolic endotoxemia and inflammation.[45-51]

Extensive exploration of the relationship between the microbiota and GVHD in animal models has shown that germ-free mice undergoing allogeneic bone marrow transplantation are resistant to the development of GVHD.[52,53] Clinical data show that GVHD involvement in the gastrointestinal tract leads to increased systemic complications and mortality rates.[54,55] Furthermore, gastrointestinal GVHD disrupts the intestinal barrier function at the intestinal microbiota/epithelial cell microenvironmental interface, eventually leading to the disruption of intestinal homeostasis.[5] Although a germ-free environment would avoid the development of aGVHD, this is an almost impossible task in humans. Moreover, intensive antibiotic treatment to achieve this goal has yielded controversial clinical outcomes.[6] Bilinski et al reported that colonization by antibiotic-resistant bacteria is associated with an increased risk of aGVHD and a higher mortality rate.[56] Mucositis caused by damage to the intestinal barrier is a common side effect of radiotherapy and chemotherapy treatments and is linked with aGVHD.[57] The microbiota themselves can also contribute to the development of mucositis via multiple mechanisms including activation of inflammatory processes and influences on intestinal permeability, mucus production, and the control of epithelial repair, as well as promoting the activation and release of immune effector molecules.[58-61] Damage to Paneth cells is one of the most recognized mechanisms of aGVHD in particular. These cells are located adjacent to intestinal stem cells within the intestinal crypts and secrete protective antimicrobial proteins such as Reg3α and defensins,[62] which mediate the selective killing of noncommensal microbes while preserving commensals.[63] GVHD-mediated inflammation depletes Paneth cells resulting in a marked reduction in alfa-defensin production and a consequent imbalance in the microbiota composition.

The advent of next-generation sequencing of 16S ribosomal RNA genes has highlighted the true complexity of the microbial gut architecture and its interactions with the host. In the case of patients undergoing aHSCT, this technology has revealed the relative contribution of different components of the microbiota to the outcome of the treatment. A significant reduction in the microbial diversity during the first week after aHSCT was associated with higher mortality and GVHD-related mortality.[38,64] During the peritransplantation period, a higher frequency of monocolonization by pathogenic microbes such as *Enterococcus* and *Streptococcus* can occur.[64] Ilett et al reported that a low level of microbial diversity is associated with aGVHD in aHSCT recipients.[65] Moreover, Peled et al confirmed that loss of microbial diversity led to the concentration of single bacterial taxa, whereas greater microbial diversity was associated with a lower risk of death in patients undergoing aHSCT.[64] These findings parallel the results of a recent meta-analysis conducted by Gavriilaki et al,[66] which showed that aGVHD alters the microbial composition by promoting Lactobacillales populations, including the genus *Enterococcus*, Staphylococcaceae, Enterobacteriales, and Pasteurellales. It has been reported that *Lactobacillus* and *Blautia* play a protective role in aGVHD, whereas *Enterococcus* plays a negative role.[67-69] In addition, depletion of Clostridia exacerbates aGVHD with a negative impact on the patient survival rate, whereas increased levels of Clostridia have the opposite effects.[70,71] Clostridiales produce 2 important SCFAs, butyrate and propionate, and inhibit the histone deacetylases, which in turn promotes the acetylation of histone H3 in Treg, thereby inducing Treg differentiation.[72] In addition, several strains of Clostridia have been reported to produce an anti-inflammatory effect by promoting the production of IL-10 in the gut.[73] In addition, Simms-Waldrip et al reported a lower level of Clostridia in children with aGVHD.[70] Miltiadous et al demonstrated that CD4+ T-cell recovery is associated with the early intestinal microbial composition post-HSCT, and also, that an increased fecal relative abundance of the genus *Staphylococcus* in aHSCT has a detrimental effect on the CD4+ T-cell recovery phase.[74] Thus, accumulating evidence supports the role for the gut microbiota and its composition in the development of GVHD by influencing immune cell activation.

Microbial metabolites also play a crucial role in intestinal immune responses and reports suggest that levels of the most common SCFAs, acetate, propionate, and butyrate, are decreased in aGVHD.[75] The cellular effect of SCFAs is mediated by activation of G-protein coupled receptors (GPRs), mainly GPR43 and GPR109A. Docampo et al reported that deletion of the GPR109A receptor in mice had not impacted GVHD, whereas transplantation of T cells from GPR109A knockout mice significantly reduced the morbidity and mortality in allo-HCT mice, confirming a link between metabolic homeostasis and T-cell activation.[76] Furthermore, GPR43 activation is important in mediating the anti-GVHD effects of butyrate and propionate, with Fujiwara et al reporting that activation of the GPR43 receptor is reduced in allo-HSCT recipients.[77] In addition, Ghimire et al demonstrated that increased GPR expression enhances a protective and regenerative immunomodulatory response, suggesting that GPR expression reduces the effects of GVHD.[78] Other than SCFAs, microbiota-derived indole and its derivative are decreased in aGVHD patients, and this correlates with aGVHD mortality.[79] Trimethylamine-N-oxide (TMAO), another microbial metabolite derived from certain dietary nutrients, such as choline, lecithin, L-carnitine, and butyrobetaine,[80] has also been found to increase the severity of aGVHD through M1 macrophage polarization in mice.[80] Furthermore, enzymes involved in the TMAO pathway, such as an inhibitor of trimethylamine (TMA) lyase, 3,3-dimethyl dimethyl-1-butanol, and hepatic flavin monooxygenases, have been reported to effectively reduce GVHD and are currently being evaluated in a clinical trial.[80]

All these observations demonstrate that the architecture of the microbiome is altered in human GVHD and implicates the gut microbiota as an aGVHD-associated biomarker. Moreover, the early microbial composition represents not only a marker of GVHD, but may also influence the condition, suggesting that exploitation of the microbiome may be a valuable therapeutic approach. This concept was pioneered in the field of oncology, with specific microbial profiles shown to be associated with different types of cancer. Moreover, some clinical trials have been conducted to evaluate the use of probiotics in combination with more traditional interventions such as immunotherapy, chemotherapy, and radiotherapy.[81-83]

MANIPULATION OF THE MICROBIOTA IN aGVHD

In recent studies, researchers have explored the potential of microbiome modulators for treating and preventing aGVHD, starting before transplantation and continuing after the procedure.[7,84] It is widely believed that modulating the host microbiota, even before transplantation, has the potential to modulate the immune response to reduce the risk of aGVHD. This concept is based on assumption that the application of targeted antimicrobial prophylaxis during the conditioning regimen, together with the use of probiotics and/or fecal material transfer (FMT) during neutropenia and the post-engraftment period, may reduce the donor T-cell activation and inflammation, reduce neutrophil activation, and improve the general effectiveness of the intestinal barrier.[84] To date, the most widely explored and encouraging option is the use of FMT, with multiple clinical trials showing partial remission rates of 74% and complete remission achieved in 50% of GI aGVHD cases, particularly in steroid-refractory patients and those with steroid-dependent disease.[85] Since FMT has no direct link with diet and has been reviewed elsewhere,[85,86] we will focus our review on pro-, pre-, and postbiotics; dietary supplements; and nutritional status.

PRECISION NUTRITION AS THERAPEUTIC APPROACHES TO aGVHD

Animal and human studies have confirmed that diet is a primary modulator of the gut microbial community.[87-89] Therefore, diet and specific nutrients are now under investigation as a new therapeutic frontier for targeting specific microbial species in both health and disease.[90] Here, we review the available studies on the effects of diet and nutritional status on the gut microbiota in patients undergoing aHSCT. We focus on probiotics and prebiotics, gut modulators that are often present in food products; we then discuss other non-prebiotic nutrients and the effect of the pretransplant nutritional status and PN and EN on the gut microbiota.

Probiotics

Probiotics are live microorganisms that are intended to have health-promoting benefits when consumed or applied to the body.[91] The most common sources of probiotics are yogurt, mature cheeses, and other fermented foods, such as kefir, kimchi, and miso, which contain mainly *Lactobacillus* and *Bifidobacterium* species.[92] The consumption should be recommended to improve the gut health and immune system functions.[93] The use of probiotics from food sources poses many technological limits in terms of factors such as the most appropriate choice of food matrix, food-processing conditions, dosage, and storage conditions.[94]

The effects of probiotics on GVHD patients have rarely been investigated for several reasons. First, many concerns have been raised around the safety of using probiotics, particularly *Lactobacillus* species, which have been linked with systemic infections in GVHD patients. In one study, although the number of infections caused by bacteria normally used as over-the-counter probiotics accounted for only 0.5% (19 of 3796) of the study population, most were caused by *Lactobacillus* species.[95] On the other hand, conflicting results were obtained in 2 studies of pediatric patients undergoing aHSCT *Lactobacillus* species administration.[96,97] Specifically, the safety and feasibility of *Lactobacillus* species administration were confirmed in both a retrospective study, in which adverse events were evaluated in 14 patients who consumed probiotics at any time before day +100 posttransplantation,[96] and an interventional study of 30 patients started with *Lactobacillus plantarum* before the transplant and monitored for 2 wk after the transplant.[97] However, in a randomized interventional study on the incidence of GVHD and the gut microbiota composition in 31 patients, *Lactobacillus rhamnosus* treatment for 12 mo had no appreciable effects on either the incidence of GHVD or the microbial composition when compared with untreated subjects and the study was terminated.[98] The variation in the results of these studies reflects the complexity of the gut microbiota and its finely tuned regulation by numerous factors. Consequently, larger studies are required to identify probiotics that modulate the microbiota composition and act on specific disease features.[99] Most of the studies evaluated formulated probiotics rather than those obtained from food sources due to the technological limits discussed above.[94] Moreover, in the case of aHSCT patients, most of these unpasteurized fermented foods are not recommended due to the risk of contamination with infectious organisms.[100] Finally, the daily dose of probiotics provided as supplements and tested clinically for their benefits in health and the majority of pathological conditions is 10^7–10^{10} colony-forming units (cfu),[94,101] which is generally higher than the levels contained in a food matrix[101]; therefore, we conclude that formulated probiotics may be a more appropriate choice for these patients.

Prebiotic Foods and Supplements

Prebiotics were first defined in 1995 by Glenn Gibson and Marcel Roberfroid as "a nondigestible food ingredient that beneficially affects the host by selectively stimulating the growth and/or activity of one, or a limited number of bacteria in the colon, and thus improves host health."[102] Plant-based foods, such as soluble fibers, yeast bread, and bulgur, are known to have prebiotic effects.[103-105]

Few studies have evaluated the effects of a prebiotic-rich diet in transplanted patients. In a Turkish study, the effect of pretransplant dietary intake on transplantation outcomes was investigated in a pediatric population. Evaluation of responses to a food frequency questionnaire designed to assess the dietary intake of 41 children in the last week before transplantation revealed a negative correlation between the day of neutrophil engraftment and the amount of soluble fiber, iron, breast milk, a traditional Turkish yeast bread, and bulgur, whereas there was a negative correlation between the duration of febrile neutropenia and intake of yogurt and onion.[106] Most of these foods (breastmilk, soluble fibers, yeast bread, bulgur) are known to have prebiotic effects.[103-105]

Other studies have been conducted on the effects of prebiotics provided to aHSCT patients as dietary supplements, such as galacto-oligosaccharides (GOSs) and fructo-oligosaccharides (FOSs). GOS supplementation has been tested in an animal study starting from a week before transplant to day +100 and it showed a reduced GVHD clinical score at day 14, a shift in the microbial community increasing butyrate-producing species that was reflected in the high level of total SCFAs and butyrate. The investigators repeated the study on genetically identical mice from a different vendor to test the effect of the baseline microbial composition and demonstrated that the 2 mouse groups responded differently to the supplementation, suggesting the importance of a patient-personalized prebiotics supplementation to improve the aHSCT outcome.[35] The GOS are currently tested in phase I clinical trial on adult patients undergoing aHSCT (www.clinicaltrials.gov: NCT04373057). Similarly, a phase I trial tested the feasibility, tolerability, and the maximum tolerated dose of FOS supplementation in 15 subjects undergoing aHSCT, with supplementation started before transplantation (day −5) and continued for a total of 21 d. From a clinical perspective, no improvement was observed in the development of GVHD, infection rate, and OS of the patients receiving dietary FOS supplementation compared with the untreated control group. Similarly, no effect was observed on gut microbial diversity, SCFA composition, and immune response in these patients. Although this study showed that FOS administration was well tolerated, many limitations were acknowledged, including a small number of patients, antibiotic administration before transplantation, and intermittent prebiotic intake.[107] In contrast, in another study, the administration of a combination of oligosaccharides and resistant starches for a period of 28 d after transplantation was more efficient in reducing the incidence of GVHD and oral mucositis as well as the duration of severe oral mucositis and diarrhea. Microbial diversity and SCFA production were also increased in the prebiotic-treated group.[36] Although there was no improvement in the rates of GVHD and infections in 22 aHSCT patients included in a retrospective study of the effect of a prebiotics mix of oligosaccharides, glutamine, and dietary fibers, a reduction in the duration of mucositis and diarrhea in addition to a decrease in weight loss and the duration of the requirement for artificial nutrition were observed.[108] A phase 2 clinical trial is ongoing evaluating the feasibility, safety, and early efficacy of a dietary supplement containing potato-based resistant starch to subjects undergoing aHSCT starting before the conditioning phase and continuing for 100 d. Results are not available yet, but the study aims to demonstrate the efficacy of resistant starch in increasing intestinal butyrate and thus reducing the rate of aGVHD (www.clinicaltrials.gov: NCT02763033).

The prebiotic property of the lactose is still debated, and it depends on the genetic background of the subjects. Lactose is normally hydrolyzed by the human small intestine, reaching only in part the colonic microflora. However, in people with reduced lactase activity, the capacity to digest lactose without adverse effects depends on the fermentation capacity of the microbiota. The lactose intestinal fermentation produces similar metabolites to the GOS, providing a certain prebiotic effect, if ingested in limited amounts. On the contrary, in the case of immature microbiota or dysbiosis, that is, due to antibiotic treatment, excessive lactose fermentation can cause pathogenic bacteria overgrowth and diarrhea.[109] Lactose intake and lactose intolerance have been associated with an increased abundance of *Enterococcus spp*. and risk of aGVHD. Stein-Thoeringer et al performed a multicentric clinical study on allo-HSCT patients to investigate the role of *Enterococcus* spp. in the development of aGVHD, in which they observed reduced OS and increased GVHD-related mortality in subjects with *Enterococcus* fecal domination in the first days after the transplant (day 0 to +21). To better understand the effect of *Enterococcus* in the onset of GVHD, in the same study, the authors created a murine model of GVHD infected by *Enterococcus* spp. The metabolic pathway analysis showed the enrichment of the lactose and galactose pathways. After demonstrating that Enterococci grow on lactose, they tested the effect of a lactose-free diet on the transplanted mice and observed a significative reduction of *Enterococcus* and improved GVHD phenotype.[110] To confirm the effect of a lactose-free diet in the patients, the authors genotyped the allo-HSCT patients for the genetic variant (rs4988235) of the lactase gene affecting lactose absorption in the human subjects.[111] The patients carrying the genetic variant, classified as lactose malabsorbers, showed a prolonged enterococcal domination in the gut. The authors concluded that a lactose-free diet in genetically predisposed patients, can attenuate the *Enterococcus* expansion after transplant and the risk of GVHD.[110]

Protein-derived supplements are also investigated for their potential prebiotic effect in improving malnutrition and muscle loss in oncologic patients. Soy-whey blended protein supplementation in HSCT patients improved the muscle mass and strength and increased the microbial diversity, upregulating beneficial bacteria among which *Ruminococcus*, *Lactobacillus*, Faecalibacterium, and *Veillonella*, and downregulating *Streptococcus* and *Enterococcus* only in patients with improved muscle mass.[112,113] These studies suggest the importance of a high-quality protein diet before and after transplant, particularly in patients at risk of developing protein-energy malnutrition and muscle wasting.

Similar to probiotics, the effects of formulated prebiotics for the treatment of aHSCT patients have been investigated more extensively than those of the foods containing the prebiotics. Therefore, despite the general beneficial effects of a high-fiber diet and fermented foods, the benefits of prebiotics in aHSCT patients remain to be fully established.

Non-prebiotic Nutrients

The potential benefits of non-prebiotic nutrients on GVHD have also been explored. These are mostly represented by vitamins and amino acids administered as dietary supplements to allo-HCST patients.

Tyrosine supplementation has been tested in a murine model of aGVHD, starting 1 wk before transplantation and 2–4 wk after. Tyrosine supplement reduced weight loss and diarrhea, and also improved intestinal permeability and skin integrity, particularly in the early stages of GVHD. Microbiome analysis revealed increased alpha-diversity and restoration of the levels of Bacteroidetes and Verrucomicrobia phyla after 28 d of supplementation as well as increased fecal levels of tyrosine and its metabolites from day 14. The general improvement in weight loss and skin and intestine epithelium integrity may be accounted for by the key role of tyrosine in protein structure. The possibility that tyrosine supplementation indirectly improves the posttransplantation immune milieu and tissue tropism highlights the potential value of single amino acids for the treatment of allo-HCST.[114]

The beneficial effects of vitamin A and retinoic acid (RA) on the intestinal epithelium were also evaluated in an animal model of GVHD. Surprisingly, compared with the untreated controls, a worse outcome was observed after 8 wk of supplementation, with increased intestinal permeability, reduced microbial diversity, and expansion of pathogenic microbes. The authors speculated that the potential RA-associated adverse reaction was due to the high inflammatory status in GVHD or to the negative feedback-induced reduction in RA signaling in the intestinal epithelium caused by the exogenous RA, although these hypotheses require further investigation.[115] Recent studies of the effects of vitamin C intake on gut microbiota have revealed benefits on cardiac function.[116] Pretransplant intake of vitamin C was found to be inadequate in an aHSCT pediatric population and the amount of diet-derived vitamin C was lower in the patients with a longer duration of febrile neutropenia posttransplantation.[106] Although these observations highlight the importance of adequate vitamin C intake before the procedure, the effect of vitamin supplementation on the intestinal barrier and gut microbiota requires further investigation.

Other vitamins and minerals are not directly linked to the status of the gut microbiota in allo-HSCT patients; however, indirect evidence suggests that maintaining an adequate intake of vitamin D, calcium, magnesium, glutamine, and omega-3 fatty acids can contribute to restoring the gut microbiota after the transplant. Reduced bone density and lower levels of serum vitamin D have been associated with sarcopenia in transplanted patients. Dietary intake of vitamin D and calcium was insufficient for 83.5% and 96.4% of the patients.[117] Pretransplant vitamin D deficiency and the expression of the vitamin D receptor (VDR) were also associated with a higher risk of aGVHD.[118,119] Multiple interventional trials supplementing vitamin D before and after the transplant showed a general improvement in HSCT outcomes[120-123] with some differences according to the VDR genotypes.[124,125] The modulatory effect of vitamin D supplementation on the gut microbiota has been investigated in other medical conditions, showing a strong effect, particularly on the phyla Bacteroides and Firmicutes, as summarized in a recent systematic review.[126] No study evaluated the effect of vitamin D supplementation on the gut microbiota of HSCT patients. However, alterations of the gut microbiota are known in subjects suffering from osteoporosis and even more evident in those with severe osteoporosis. It is suggested that the dysbiosis, first affecting the intestinal adsorption, may explain the lower levels of serum vitamin D, and then indirectly modulate the bone metabolism.[127,128] We can speculate that these mechanisms are common also to allo-HSCT patients suffering from low bone density and that the benefits of vitamin D supplementation are mediated by the gut microbiota. Instead, the immunomodulatory effect of vitamin D has been investigated in HSCT patients. A retrospective study found that vitamin D deficiency at day 30 after transplant was more common in lymphoid malignancy and subjects undergoing PN and myeloablative conditioning. Only skin aGVHD risk was higher in vitamin D-deficient patients, without difference in OS and cGVHD. Gene expression analysis identified perturbations in the epigenetic regulation of inflammatory genes and T-lymphocyte proliferation. Sufficient levels of vitamin D were able to reduce the T-cell proliferation via epigenetic mechanisms.[129] A multicentric prospective study evaluating the effect of day vitamin D supplementation (high and low doses versus not receiving vitamin D) from 5 d before transplant to day +100, showed a reduced incidence of cGVHD but not aGVHD and a lower number of circulating B cells and interferon at day +100 with both low and high doses of vitamin D.[123] More studies are required to explicit the molecular mechanism of vitamin D modulation of the gut microbiota in HSCT patients.

A recent study showed that optimized calcium: magnesium intake ratio improves the microbial metabolism of medium-chain fatty acids and reduces circulating levels of sucrose, minimizing the detrimental effects of sugars on metabolic traits.[130] Magnesium supplementation is recommended in GVHD patients when deficient.[29] The intake of omega-3 fatty acids shows beneficial effects as well for GVHD[29,131] and clinical trials testing omega-3-rich foods and supplements in healthy subjects showed beneficial effects on the gut microbiota and the lipids profile, increasing SCFA-producing bacteria.[132-134] Finally, glutamine is usually supplemented to allo-HSCT patients, and, as mentioned above, it reduces mucositis, diarrhea, and the duration of EN, and it improves weight loss in combination with the prebiotic oligosaccharides and dietary fibers.[108] In obese subjects, glutamine supplementation reduces the abundance of the phyla Firmicutes and the ratio Firmicutes/Bacteroidetes usually associated with inflammation, suggesting an anti-inflammatory role of glutamine acting via the gut microbiota.[135] In animal models of colon cancer, glutamine showed a protective effect on the gut microbiota against the chemotherapy, preserving species from the Bacteroides, *Lactobacillus*, *Clostridium*, and Enterobacteriaceae groups.[136] We can assume that similar microbial mechanisms can explain the beneficial effects of glutamine in allo-HSCT patients.

Postbiotics

The International Scientific Association for Probiotics and Prebiotics defined postbiotics as “relating to, or resulting from living organisms”, with the prefix “post” (meaning “after”) added to signify that this category comprises nonliving organisms more generally referred to as microbial metabolites.[137] SCFAs, which are the products of the bacterial fermentation of dietary fibers, have become a focus of interest among researchers due to evidence of their pivotal immunomodulatory properties.[138] In the context of GVHD, studies in both humans and animals have revealed that a reduced level of SCFAs (particularly butyrate) is associated with antibiotic treatments and shrinkage in the population of butyrate-producing microbes.[71,139] Restoration of butyrate levels leads to a reversion of the deficit phenotype and improves intestinal function as well as downregulation of apoptotic gene expression and upregulated junctional protein expression via epigenetic mechanisms.[71] As discussed previously, intervention trials of a mix of oligosaccharides and resistant starches, as well as PE-EVOO have provided evidence that prebiotics and probiotics increase the production of SCFAs, with beneficial effects in aHSCT patients.[31,107,108] Alternatively, SCFAs can be administered as postbiotics and have been investigated in this context for their ability to correct microbiome dysbiosis in inflammatory disorders.[140,141] SCFAs have multiple effects on the immune system, including anti-pathogenetic properties, lipids metabolism support, anti-inflammatory characteristics, and proapoptotic capacity against cancer cells.[142] Dietary postbiotic supplementation in healthy suckling rats showed beneficial effects on immune system maturation, intestinal barrier function, and the composition of the microbiota and its metabolites. These effects on animal growth were synergistically enhanced by the combination of post- and prebiotics, suggesting that this approach has the potential to modulate the immune system and microbiota starting in early life.[143] SCFAs are also known to alter gene expression in Treg cells by inhibiting the activity of histone deacetylase (HDAC),[144] which functions as a transcription repressor to regulate gene transcription by activating chromatin condensation. HDAC inhibitors (HDACis) are used as antitumor agents[145] and are known to reduce inflammation in the gastrointestinal tract[146,147]; therefore, HDACi, in particular butyrate, has been assessed for its potential to reduce gastrointestinal GVHD.[148,149] Butyrate levels have been found significantly lower at day 14 after transplant and butyrate administration was able to restore the intestinal epithelial functions and the gut microbiota in a mice model of GVHD.[71] Indeed, the promise of this approach was highlighted by a decreased GVHD incidence in a phase 2 clinical trial (www.clinicaltrials.gov, NCT02588339)[150] and larger clinical trials are testing HDACi in aHCST (www.clinicaltrials.gov, NCT02763033). Thus, we speculate that SCFAs can be used as an alternative to HDACis to modulate HDAC activity, although this hypothesis remains to be confirmed in more human clinical trials. Postbiotics recently raised interest for their safety of use, not containing any living cells, and for the multiple beneficial effects on the intestinal barrier and immune system.[151,152]

Other microbial metabolites have been investigated for their capacity to modulate the immune system in GVHD. Dietary choline, phosphatidylcholine, and carnitine are degraded to TMA from intestinal bacteria, and subsequently converted into TMAO in the liver. A murine GVHD model treated with TMAO or choline starting 2 wk before transplantation and continuing for 50 d after, showed a deterioration in GVHD scores for both treatments, whereas the choline analog, dimethyl-butanol (DMB), reversed the deleterious effects of choline. The authors also demonstrated that TMAO stimulates GVHD progression through the activation of M1 macrophages and Th1 and Th17 lymphocytes but were unable to clarify the link between dietary choline, TMAO, gut microbiota, and GVHD. Nevertheless, they speculated that elucidation of the underlying mechanism would provide a basis for the development of new interventions for the control of GVHD by modulating the microbiota and its metabolites.[80] The dietary sources of choline, phosphatidylcholine, and carnitine are eggs, liver, dairy products, peanuts, etc, mostly from animal proteins. Since plant protein foods, such as soy, showed a protective effect versus GVHD,[112,113] we can speculate that the quality and the source of dietary proteins may affect the HCST outcomes, modulating the microbiota. Therefore, the dietician should carefully balance the protein intake during the nutritional intervention pretransplant.

EN and PN

Nutritional status and nutritional interventions also influence the composition of the microbiota and SCFA levels. The previously mentioned meta-analysis supporting the benefits of EN compared with PN in aHSCT indicates a potential anti-inflammatory and immunomodulatory mechanism that reduces microbial molecule translocation and increases the production of SCFAs.[18] A longitudinal study measured the effect of EN versus PN on the gut microbiota in a pediatric cohort of HSCT patients (EN = 10, PN = 10). The microbial composition and metabolites did not undergo a significant shift for EN versus PN after the transplant, with a faster recovery of beneficial species, such as *Faecalibacterum*, *Dorea*, *Blautia*, Bacteroides, Parabacteroides and *Oscillospira*, and consequently, an increased level of the SCFA in subjects using EN.[153] A randomized controlled trial of the effects of EN versus the clinical standard (including PN) for 30 d from the transplant on the gut microbiota showed no differences in the alpha diversity, although patients under EN showed a higher abundance of Bacteroidetes and a lower abundance of Proteobacteria. SCFA-producing species were particularly increased under EN. Moreover, prolonged oral intake was shown to improve microbial diversity and SCFA-producer species. However, it should be noted that also this trial was performed as a pilot study with a small number of subjects (EN = 12; PN = 11) and the conclusions require confirmation in larger randomized controlled trials.[154] The composition of the EN formulas also affects the microbial composition and the fiber content needs to be carefully evaluated during the nutritional intervention. The Food And Resulting Microbial Metabolites study showed that EN devoid of fiber delays the microbial restoration after antibiotic treatment, compared with both omnivore and vegan diets in a healthy volunteers trial. In the EN group, the microbial composition and metabolites were characterized by predominant Proteobacteria versus Bacteroidetes and Firmicutes, reduced production of carbohydrates metabolites (such as butyrate), and altered amino acids metabolism (such as tryptophan). This study demonstrated the importance of fiber content in the nutritional intervention to restore healthy microbiota after antibiotic treatment.[155]

Pretransplant Nutritional Status

Currently, there is no direct evidence of the role of the gut microbiota in malnutrition after aHSCT; however, the emerging literature shows that the microbiota can modulate muscle metabolism and the development of sarcopenia in the elderly population.[156] The potential mechanism of these effects is based on the balance between beneficial and harmful bacteria that influence the skeletal muscle mass, thereby controlling the inflammatory status and insulin resistance. Endurance exercise and a balanced diet (adequate intake of protein and fibers) can help to maintain muscle mass in the elderly population.[156] Similarly, a recent study showed that healthy eating and an active lifestyle can also help prevent severe malnutrition in cancer patients through the effects on the gut microbiota. Despite these revelations, the mechanisms underlying the effects of different dietary patterns on the gut microbiota in both tumorigenesis and cancer treatment (ketogenic diet, fasting-mimicking diet, high-fiber intake) remain to be clarified.[157]

As discussed, the effect of obesity on HSCT outcomes remains controversial.[15,21,24-27] Obesity has been associated with less diverse microbiota with more pronounced proinflammatory profiles.[158] To date, only murine models of diet-induced obesity have revealed the involvement of gut microbiota in poor GVHD outcomes.[22,23] In particular, a high-fat diet reduced microbial diversity that was associated with GVHD and induced an irregular microbial profile comprising an increased abundance of *Akkermansia muciniphila* and *Enterococcus*, and a reduced abundance of Clostridiaceae family species.[22] These data clearly show that controlling body weight can improve aHSCT outcomes through microbiota modulation in a manner similar to that observed in many chronic disorders in which obesity increases the inflammatory status and the risk of complications.[159] Therefore, healthy eating habits and an active lifestyle should be strongly encouraged in the general population to contribute to reducing the obesity-related risks for both acute and chronic diseases, such as aGVHD.[22,23,160] Thus, the current evidence warrants further exploration of the role of obesity in microbiota regulation in aGVHD patients.

CONCLUSIONS

In this review, we have discussed some of the many studies that have been conducted to clarify the role of the gut microbiota in aHSCT and GVHD as well as the potential therapeutic approaches (summarized in Table 1). Although support for the therapeutic benefits of nutrients, prebiotics, and probiotics is limited, there is very robust evidence that FMT is a therapeutic valid option in gut GVHD, even in the most severe forms.[85,86] This evidence confirms that the microbiome has a key role in modulating inflammatory gut conditions typical of some cases of aHSCT-related GVHD. The limitations of the findings emerging from the studies of single microbiome modulators are most likely due to the lack of knowledge about the immune mechanisms that regulate the gut microbiota composition. Importantly, in the era of personalized therapy, the characterization of the microbiota of each patient will facilitate more precise and effective intervention, thus improving outcomes.[161]

FUTURE PERSPECTIVES

Having highlighted the importance of nutritional status and the capacity of dietary components to modulate gut microbiota in HSCT patients (summarized in Table 1), we propose that a better scientific understanding of the immunological properties of the human microbiome will open up new avenues for the development of targeted treatments that reduce the risk of GVHD. Furthermore, therapies could be designed to select for a protective commensal flora and eliminate pathogenic bacteria in a 3-fold fashion: (1) a selected diet and prebiotics directed at modifying the microbiota in favor of GVHD prevention; (2) selecting specific probiotics and postbiotics to promote the expansion of the beneficial microbes and increase the levels of their metabolites; and (3) strategizing the prophylactic antimicrobial therapy before and during aHSCT to redirect the microbiota toward a GVHD protective function. Future research initiatives will be required to optimize these methods and select the most appropriate one for each situation. Moreover, since enhanced effects have been achieved by a combination of >1 therapy,[108,139] this approach also warrants further investigation. However, the personalization of these interventions is dependent on the precise characterization of the patients in a stepwise manner. Before transplantation, investigations should focus not only on the gut microbiota composition, but also on nutritional status, lifestyle, dietary habits, and medications. After transplantation, changes in the gut microbiota should be evaluated for comparison based on the conditioning regimen and the nutritional intervention, finally measuring clear endpoints, such as time to

TABLE 1.
The effects of dietary factors and nutritional status on the gut microbiota of aHSCT patients

Diet and nutritional factors	Benefits for aHSCT	Benefits for aGVHD	Effect on gut microbiota in GVHD	Type of study	References
Probiotics (Lactobacillus species)	Good safety and acceptability	No effects	No effects	Human studies	96-98
Prebiotics					
Prebiotic food (soluble fiber, breast milk, yeast bread, bulgur, onion)	Reduction of the neutrophil engraftment days and the duration of the febrile neutropenia	NA	NA	Human observational study	106
GOS	NA	Reduced GVHD risk score on day 14	Increase of butyrate-producing bacteria; Increased level of SCFAs	Animal study	35
FOS	Good acceptability	No effects	No effects	RCT	107
Oligosaccharides + RS	NA	Reduced rate of GVHD, oral mucositis, and diarrhea	Increased microbial diversity and SCFAs production	Human study	36
Glutamine and GFO	Reduction in weight loss and artificial nutrition duration; reduction in diarrhea and mucositis duration[108]; improvement in the enterocyte metabolism[30]	No effect on GVHD rate and infections	Reduction of microbial translocation and pathogens[108]	Human study[108]; meta-analysis of human studies[30]	30,108
Lactose-free diet	NA	Improvement of GVHD phenotype	Reduction of the *Enterococcus* spp. Particularly in genetically lactose malabsorbers.	Animal study	110
Soy-whey blended protein	Improved muscle mass and strength[113]	NA	Increased diversity; upregulation of beneficial bacteria (*Ruminococcus*, *Lactobacillus*, *Faecalibaterium*, etc) and downregulation of *Streptococcus* and *Enterococcus*[112]	Human studies	112,113
PE-EVOO	Improvement of intestinal histopathology; reduced activation of T cells and inflammation	Reduced risk of GVHD	Restoration of butyrate	In vitro and in vivo animal study	31
Postbiotics					
Butyrate	NA	Decreased GVHD incidence	Gut microbiome restored on day 14 after the transplant	Animal study	71
Choline analogs	NA	Improvement in GVHD scoring, decreased ulcerations and inflammation	Reduction of TMAO metabolite	Animal study	80
Tyrosine	Reduction of weight loss and diarrhea	Improvement of the intestinal permeability and skin integrity in early-stage GVHD	Improvement of the microbial diversity; restored levels of the Bacteroidetes and Verrucomicrobia phyla	Animal study	114
Vitamin A	NA	Negative outcomes: increased intestinal permeability	Negative outcomes: reduced microbial diversity and expansion of pathogenic microbes	Animal study	115
Vitamin C	Inadequate amount associated with longer febrile neutropenia	NA	NA	Human study	106

Continued next page

TABLE 1. (Continued)

Diet and nutritional factors	Benefits for aHSCT	Benefits for aGVHD	Effect on gut microbiota in GVHD	Type of study	References
Vitamin D	HSCT outcomes improvements,[120-123] according to VDR genotype[124,125]	Higher GVHD risk in vitamin D-deficient patients.[118,119] Vitamin D supplementation before and after the transplant reduces cGVHD incidence[123]	NA	Human studies,[118,120,125] not randomized clinical trials[119,122-124] and RCT[121]	118-125
Omega-3	NA	Improve GVHD condition and reduce inflammatory biomarkers	NA	RCT	131
EN	Improvement in the intestinal mucosa, reduced incidence of hyperglycemia and infections compared with PN[18]	Reduced risk of GVHD	Increased abundance of SCFA-producer bacteria, anti-inflammatory and immunomodulatory effects[153,154]	Meta-analysis of human studies[18], RCTs[153,154]	18,153,154
Obesity	Contrasting data: increased mortality and infections[15,21,27], no increased OS and infections[24-26,28]	Contrasting data: increased risk of GVHD[15,27], no changes in GVHD risk[24,25]	Reduction of microbial diversity; increased abundance of *Akkermansia muciniphila* and *Enterococcus* and reduction of Clostridiaceae family[22,23]	Human[15,21,24-28] and animal studies[22,23]	15,21-28

aGVHD, acute graft-vs-host disease; aHSCT, allogeneic hematopoietic stem cell transplantation; EN, enteral nutrition; FOS, fructo-oligosaccharides; GFO, glutamine, fiber, and oligosaccharides; GOS, galacto-oligosaccharide; GVHD, graft-vs-host disease; HSCT, hematopoietic stem cell transplantation; NA, not available; OS, overall survival; PE-EVOO, polyphenolic extracts in extra virgin olive oil; PN, parenteral nutrition; RCT, randomized controlled trial; RS, resistant starch; SCFA, short-chain fatty acid; TMAO, trimethylamine-N-oxide; VDR, vitamin D receptor.

mucosal integrity reconstitution, resolution of mucositis side effects, and the development (or not) of GVHD. Maintaining healthy microbiota and nutritional status will clearly reduce the risk to develop GVHD; therefore, the scientific community is called to increase efforts in developing precise therapies focusing on the use of probiotics, prebiotics, and postbiotics for the prevention and management of GVHD.

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